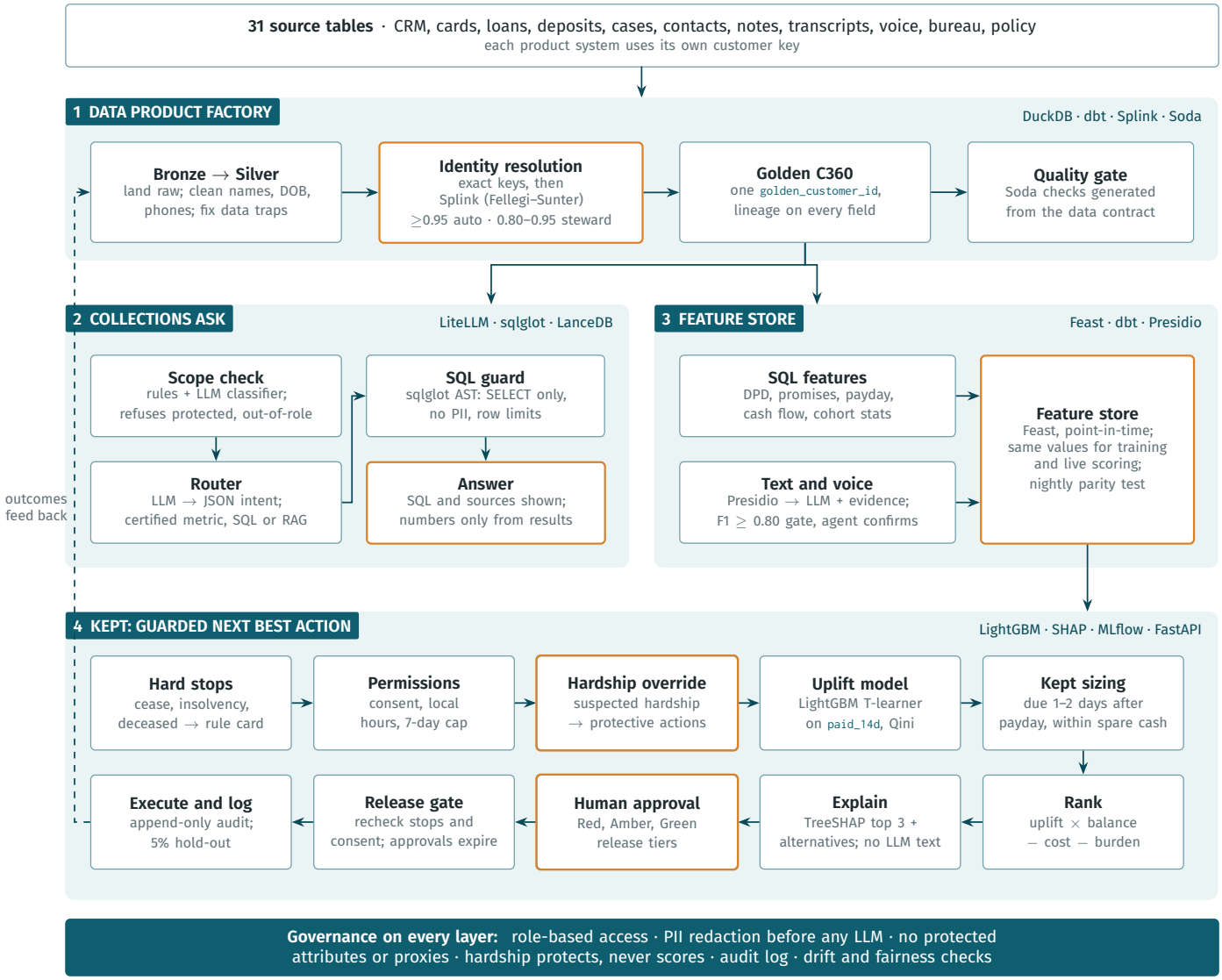
Kept: a guarded next best action for promises customers can keep

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WHAT THE AGENT SEES — Case C-118204 · G-004817 · nba-0.3 · features as of 28 Sep 06:00 · illustrative

Call today 18:00–20:00: check circumstances first, then offer a 3-month plan, $140 due 3 Oct (the day after payday)

Chance of paying or keeping the promise in 14 days: **Plan call 58%** | Hardship review 47% | Wait 22%

Why	Pay lands on the 2nd; the last two promises fell due before payday and broke	**+9 pts**
	Pay is down 27% on the 3-month average	**+6 pts**
	Answered 4 of the last 5 calls after 18:00	**+4 pts**
Kept sizing	$140 fits inside estimated spare cash of ~$160 a month after rent; due one day after the expected payroll date.	
Similar cases	812 past cases like this: 64% kept a payday-dated plan vs 39% a lump-sum promise (descriptive, not causal).	
Checks	Call consent ✓ 2 of 3 contacts this week ✓ inside 07:00–21:00 local ✓	

Hardship: possible. Note 18 Sep: “shifts cut at the plant since Aug” (unconfirmed). SMS link and settlement removed; if confirmed, routes to a hardship specialist (HP-REDUCE v2 eligible).

Accept | Edit amount or date | Alternative: hardship review | Override (reason required)

1. Summary

A collections customer's history sits in 31 tables, and cards, loans, deposits and CRM each use their own customer key. Agents see one slice of it, so treatment is generic and often lands before payday.
We build *Kept*: an NBA card that picks the action, channel and time most likely to stop a case rolling, and dates and sizes any promise around the customer's payday. Safety rules run first, and a person releases every action.
Users: collections agents and hardship specialists, team leads, strategy analysts (through Collections Ask), and compliance.

2. Business Value

We track one number: the **30-day roll-forward rate** for cases entering at 1–30 days past due (share of cases where `dpd_at_day_30` lands in a worse bucket than `entry_dpd`). We also watch the promise-kept rate, and we don't let complaints, late hardship detection, override rate or cost per dollar recovered get worse.

Two things should move it. We only contact people when contact changes the outcome (uplift), and we date promises to payday using `expected_next_payroll_date`, which today's strategy ignores. Our hypothesis is 1–2 points lower roll rate at the same or fewer contacts. We'll put a dollar figure on it on build day 1 (roll reduction $\times$ monthly inflow balance $\times$ charge-off share).

To measure it, we compare against the current strategy (`nba_recommendation_last`) on the randomly assigned `test_cell` rows using Qini curves, then keep a permanent 5% hold-out on the old strategy and read it at day 30.

3. Layer 1: Data Product Factory

Everything runs on DuckDB with dbt (bronze, silver, gold), which handles ~1M customers on one laptop. Bronze keeps raw files as they land. Silver cleans names, mixed date-of-birth formats, phones and postal codes, and quarantines rows it can't parse.

We also fix the traps in this dataset: account tables have one row per snapshot (we take the latest), card `cycles_delinquent` counts cycles rather than days (we use `dpd`), bureau scores use different scales (we convert to percentiles), and `transactions` only covers a sample (missing stays missing).

Matching customer IDs. First we apply exact links: case reference to CRM ID, loan and card payment accounts to deposit accounts, and identical `national_id_hash`. Then Splink scores the remaining pairs on name, date of birth, phone and postal code. Scores of 0.95 and above link automatically, 0.80–0.95 go to a data steward, and anything lower keeps a provisional ID. Clusters with two different national IDs or more than 12 keys always go to review. Each person gets a permanent `golden_customer_id`, and merges and splits are logged so any link can be undone. We target 98% precision on a 300-pair reviewed sample.

The gold layer publishes the C360 tables plus `c360_restricted`, which holds protected attributes and is never joined into features. Consent takes the most restrictive value across duplicates. An LLM drafts source-to-target mappings and schema-drift fixes as pull requests, and an engineer approves them.

4. Layer 2: Collections Ask

SQL computes the answer and the LLM only explains it. A question goes through five steps:

1. **Scope check.** Rules plus an LLM classifier refuse questions about protected attributes, bulk personal data, write actions, other people's cases, or data we don't have.
2. **Router.** The LLM turns the question into JSON (metric, filters, dates), shown as editable chips.
3. **Answer path.** Metric questions use certified definitions from `metric_definitions`, so the SQL is compiled, not written by the LLM. Lookups get three LLM-written SQL candidates, and we keep the majority result. Policy questions search current documents in LanceDB and cite the section.
4. **SQL guard.** sqlglot allows SELECT on approved tables only, blocks protected and personal columns, and adds row limits and a timeout.
5. **Grounding.** Low confidence or disagreeing SQL gets one clarifying question, then a refusal.

A refusal says why: *"This needs gender_code, which is excluded from collections analytics. Try: [nearest allowed question]."* Every answer shows the SQL, the tables used and data freshness. We score it on the provided benchmark questions (80%+ accuracy, 95%+ refusal recall) and rerun the scorecard on every prompt change.

5. Layer 3: Feature Store

Each feature is written once in dbt SQL and served by Feast. Training reads values as they were at each past decision, and live scoring reads the same values from the online store. A nightly parity test blocks deploys if the two differ by more than 0.1%, and a deny-list keeps outcome fields out of training.

Feature	Source
Max DPD, times 30+ DPD in 12 months, days in bucket	C360 accounts, collections cases
Broken promises (90 days), kept-promise ratio	promises to pay
Days to next payroll, pay frequency, payroll delay	deposit accounts
Cash-flow trend, card utilisation trend	deposit and card history
Answer rate, payment-link usage, contacts in last 7 days	contact history
Recovery and kept rates by product and bucket (groups of 50+)	cohort stats
Hardship signal (none / possible / clear)	LLM on notes and calls
Promise intent, promised date and amount	LLM on calls
Dispute, cease request, third party present	LLM on notes and calls; feed the hard stops

Text features work like this. Presidio strips personal details, the LLM returns each value with the quote that supports it, and a small SetFit classifier trained on LLM labels fills in the 1.5M historical notes. A feature ships only if it scores F1 ≥ 0.80 (hardship recall ≥ 0.90) on 300 notes we label twice. The agent confirms or edits extracted values at wrap-up. Suspected hardship restricts actions straight away, but only a person can clear it. Stated reasons like medical or bereavement only trigger hardship handling and never reach the model.

6. Layer 4: Models and Decisioning

We picked NBA because it moves roll rate directly and the data already has random test cells, the current strategy's choices, contact costs and channel capacity. There are eight actions: wait, email, push, SMS payment link, call, promise request, payment plan and hardship review.

1. **Hard stops.** Cease, insolvency, bankruptcy or deceased flags mean no collection action, and the agent sees which rule fired.
2. **Permission gate.** Drops any action without channel consent, outside the customer's local calling hours, or over the 7-day contact cap.
3. **Hardship override.** Any hardship or vulnerability signal limits actions to a check-in call, hardship review, payment plan or wait. This sits outside the model.
4. **Uplift model.** One LightGBM model per action predicts the chance of payment within 14 days, trained on randomly assigned cases and checked with Qini on the hold-out.
5. **Kept sizing.** Promises fall 1–2 days after expected payday, and the amount stays within estimated spare cash. This step is plain arithmetic.
6. **Ranking.** Uplift $\times$ balance at risk, minus contact cost and burden, within daily channel capacity. So a smaller plan that gets kept beats a bigger promise that breaks.
7. **Explanation.** TreeSHAP gives the top three reasons, mapped to fixed sentences, plus the alternatives and a "similar past cases" line. The LLM writes none of it.

Tier	Applies to	Who releases it
Blocked	Any hard stop	Nobody (compliance can override with a reason)
Red	Confirmed or unconfirmed hardship, payment reductions, settlement, LLM vs agent disagreement	Hardship specialist plus a second person
Amber	Calls, promises and plans; close calls between top two actions	The handling agent: accept, edit, pick another, or override
Green	Template reminders and SMS links	Team lead approves the daily batch after sampling 20 cases

The model never sends anything itself: a proposal is a record that a person approves, and the release step rechecks consent on fresh data. Errors or stale data mean "no action", a kill switch reverts to the old strategy, and new model versions need sign-off in MLflow.

7. AI Usage

Where	What the AI does	What keeps it in check
Layer 1	LLM drafts mappings and drift fixes; Splink matches customers	Engineer approves; steward reviews unsure matches
Layer 2	LLM scopes, routes, writes lookup SQL, narrates	Certified metrics, SQL guard, numbers only from results
Layer 3	LLM extracts hardship and intent from text; Whisper transcribes voice	Redaction first, F1 gate, agent confirms
Layer 4	LightGBM uplift picks the action; SHAP explains it	Rules before and after, human release, hold-out

Rules, consent, hardship eligibility, promise sizing and sending use no AI. The LLM sits behind LiteLLM, so we can swap a hosted model for a local one, and every call is logged.

8. Scalability

Nightly by 06:00 ET, we rebuild the C360, rerun matching on changed rows, refresh features and score every open case. After each contact, extraction and re-scoring finish in seconds, and payments or cease requests cancel pending proposals. The agent's card loads in under 300 ms, and Ask answers in under 5 s. In production the same dbt and Splink code moves to Spark or Snowflake, with Redis and Kafka.

9. Governance

Agents see only their own queue with personal data masked, analysts see aggregates of 10 or more, and only compliance can read protected attributes (for fairness audits). Every query, recommendation, override and LLM call goes to an append-only audit log.

The 12 protected attributes in the dictionary stay out of every decision, along with proxies like age, postal code, language, benefit flags and voice. A CI test fails the build if a feature comes from any of them, and Fairlearn checks recommendation rates across groups.

Hardship can come from the agent, an account programme, the LLM or a cash-flow rule (late payroll, salary drop, new EI deposits). It only ever removes actions or routes the case to a specialist, so it can never count against a customer. DNCL and CASL cover marketing, not collections, so they block marketing use of the C360 and leave lawful collections contact alone.

10. Build Plan (36 Hours)

Hours	Data and IDs	Ask	Text features	Model and UI
0–3	Load tables, contract	Metrics, benchmark harness	Redaction, label 300 notes	Label SQL, rules, app skeleton
3–10	Cleaning, Splink	Router, SQL guard	Extraction prompt and eval	Hard stops, dbt features
10–18	Gold C360, quality checks	Answer card	LLM labels, SetFit backfill	Feast, uplift model
18–26	Steward review screen	Policy search	Live extraction	Kept sizing, ranking, SHAP
26–32	Parity test	Full benchmark	Fairness tests	NBA card, release gate
32–36	Everyone: end-to-end demo (raw keys to C360 to Ask to a new note flipping hardship to the NBA changing), video and deck			

If we fall behind, we drop the semantic layer first, then voice, then the similar-cases line. We won't cut ID matching, refusals, the safety rules or the release gate.

Draft Data Contract: The Golden Customer Record

This contract covers `gold.c360_customer` and `gold.identity_crosswalk`, one row per `golden_customer_id`. It lives as an ODCS YAML file in our repo, CI lints it, and Layers 2 and 3 read their allow-lists and permitted features straight from it.

Owner	Head of Collections Strategy (business); Collections Data Engineering (technical); Collections Data Steward (identity review)
Version	1.0.0. Breaking changes get 14 days' notice
Location	DuckDB `gold` schema, plus a Parquet export partitioned by `as_of_date`
Refresh	Nightly, published by 06:00 ET. Consent, cease and hold changes land within 15 minutes through the online store

Column	Type	What it holds	Sensitivity
golden_customer_id	string, key	Permanent ID; merges and splits are logged	Internal
source_keys	list	CRM, card, loan, deposit, collections and bureau keys	Internal
match_confidence_min, match_review_status	decimal, enum	Weakest link in the cluster; auto, steward-approved or unlinked	Internal
display_name, primary_phone_e164, email	string	Most recent verified CRM values	Direct PII (masked)
province_code, time_zone, preferred_language	string	From CRM; language only for routing	Quasi-identifier
total_overdue, max_dpd, current_bucket	number, enum	Across current account snapshots	Internal
open_case_id, assigned_agent_id	string	The open collections case	Internal
contact_permissions	struct	Consent per channel, most restrictive across records	Internal
do_not_contact, collections_hold	bool	Internal do-not-call or cease; insolvency, bankruptcy or deceased	Sensitive
hardship_status, hardship_source	enum	Combined agent, account, confirmed LLM and cash-flow signals	Sensitive
expected_next_payroll_date, payroll_frequency_detected	date, enum	From deposits; drives Kept sizing	Internal
as_of_date, refreshed_at, <attr>_source	date, ts	Lineage for every field	Internal

Quality rules (Soda, every load):

1. **Unique IDs.** golden_customer_id is never null or duplicated, and each source key maps to exactly one golden ID. Fail: block.
2. **Linkage.** Every open case resolves to a golden ID, and at least 98% of account keys link. Fail below 97%: block.
3. **Match safety.** Auto links need a score of 0.95 or more, and no cluster can hold two national IDs. Fail: block. A weekly reviewed sample must stay at 98% precision or we warn.
4. **Contact safety.** Any source do-not-call or cease flag sets do_not_contact, and any insolvency, bankruptcy or deceased flag sets collections_hold. Zero violations allowed.
5. **Freshness.** Published within 26 hours of the source extract (warn), blocked at 48 hours. Row count within 10% of the last load.

Allowed uses. Collections servicing and contact, the NBA and other collections models (with human release), Collections Ask (masked, aggregated for analysts), hardship identification, regulatory and complaint reporting, and model monitoring.

Not allowed. Marketing or cross-sell (that's where DNCL, CASL and marketing opt-outs apply), credit or pricing decisions, any decision using protected attributes, bulk export of personal data, sharing outside the bank, and joining to c360_restricted outside compliance.

Still TBC. Whether autopay_source_account and bank_subject_ref work as link keys, the certified metric_id for roll rate, and whether the organisers have ground-truth IDs to score our matching.

AI assistance disclosure. We used Claude (Anthropic) as an assistant while preparing this design, for literature review, drafting and formatting. Our team reviewed, edited and approved every part of it, and all design decisions are our own.